

Lab Experiment-16

16. Implement a Sobel algorithm using Open CV to filter the input image.

Code:

```
!pip install opencv-python
import cv2
from google.colab import files
from google.colab.patches import cv2_imshow
f = files.upload()
file = list(f.keys())[0]
img = cv2.imread(file, 0)
sobelx = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3)
sobely = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)
cv2_imshow(img)
cv2_imshow(sobelx)
cv2_imshow(sobely)
```

Output:



